

How We Tested The Drives

On our test rig, the latest versions of the hardware drivers were installed along with DirectX 9.0c. The internal drives to be tested were connected to the third SATA connector of the motherboard, and the external drives were connected to the first USB port.

Test Rig

Processor	Intel Core 2 Extreme X6800 @ 2133 MHz
Motherboard	Gigabyte GA-965G-DS3
Chipset	Intel 965G
RAM	1 GB Kingston HyperX DDR2 @ 533 MHz
Host hard drive	Seagate Barracuda 750 GB SATA II
OS	Windows XP with SP2

The Test Procedure

The performance tests of the hard drives consisted of synthetic tests using benchmarking software and real-world tests where application performance and file transfer speeds were tested.

The synthetic tests were conducted using two types of benchmarks: the low-level benchmarks HD Tach RW 3.0.1.0 and H2Bench, and standard benchmarks PCMark05 and SiSoft Sandra Pro Business 2007, all of which are popular benchmarking software for storage devices.

Features

We noted features of the drives such as capacity, dimensions, weight, bundled software, interface type, cables and accessories, ruggedness, and more.

Performance

We gauged performance based on eight tests, as follows.

Low-level Synthetic Benchmarks: HD Tach RW 3.0.1.0

The hard drive to be tested was connected to the test machine, and was not formatted. It was allowed to remain raw (unpartitioned). HD Tach RW 3.0.1.0 was then run and used to evaluate the drive. The scores we noted were as follows:

CPU Utilisation: This benchmark determines how much load the device puts on the CPU. Lower CPU utilisation is better. High CPU utilisation (above 15 per cent) generally indicates that a poor controller is being used, that DMA needs to be enabled, or that a driver update is required.

Random Access Time Benchmark: This determines the random access speed of the device. Random access is the average time it takes to retrieve data from a randomly located sector on the device. Lower random access speeds mean better application and database performance. Random access times can also give an insight into the efficiency of the interface a device is attached to.

Sequential Read and Sequential Write: The maximum sustained speed is an important metric for real-time sequential access applications. This maximum speed may be important to a power user who would like to partition the drive into a high-speed partition and a low-speed partition.

Burst speed: This represents the limits of the throughput that a drive is capable of delivering.

Low-level Synthetic Benchmarks: H2Bench (Internal Drives Only)

This is a new benchmark developed by Heise Zeitschriften Verlag GmbH & Co. KG, which we used for the first time in a hard drive test; it provides very accurate results. This is a Win32 program that needs to be run from the command line. The interface speed at 50 per cent of the drive capacity was noted. The "Core Test" or repetitive sequential read test was then performed, and this gave the maximum transfer rate the drive can achieve after repeated sequential reads. The read and write seek times were also noted.

Standard Synthetic Tests

A 32 GB NTFS partition was created where possible with compression turned off on the test hard drive. A second partition was created, utilising the remaining capacity.

PCMark05 (Internal Drives only)

FutureMark's PCMark05 is a very popular tool to benchmark the entire system. This benchmarking software has different modules that can be chosen to benchmark a particular PC component. We installed Windows XP SP2 on the test hard drive with the latest drivers, as on the test rig, and installed and ran this benchmark application. We noted the scores of the following components from the hard drive benchmark module.

a. XP Startup: This simulates activity of Windows XP startup and tests the performance of the hard drive at that time.

b. Application Loading: This simulates opening and closing of several applications such as Adobe Acrobat Reader, Windows Media Player, and more.

c. Virus Scan: This tests drive performance during a simulated virus scan of approximately 600 MB of files of different types.



RPMs And Buffers

The Western Digital Raptors whirled away at rpms as high as 10,000, which was the highest amongst all the drives. This contributed to the performance of these drives during the actual tests. The rest of the drives had a stated rpm of 7,200.

Buffer sizes ranged from 8 to 32 MB. Hitachi's 1 TB behemoth was the only drive with a huge 32 MB buffer, which should theo-

retically help it when big file transfers occur, but in the tests we conducted, there wasn't a significant gain. All the Western Digitals and Maxtors featured 16 MB buffers, as also the Hitachi Deskstar 7K500, Seagate Barracuda 7200.9 ST3500641AS, and the Barracuda 7200.9 ST3750640AS. The lowest buffer was 8 MB, and drives with this amount were nearly all from Samsung. However, in everyday operations, even this amount of buffer is sufficient.